

SQL Queries For Mutual Fund Analysis

-- Query 1. Top 5 Funds by AUMSELECT

```
f.scheme_name,
p.aum_crore
FROM fact_performance p
JOIN dim_fund f
ON p.amfi_code = f.amfi_code
ORDER BY p.aum_crore DESC
LIMIT 5;
```

	scheme_name character varying (250) 	aum_crore numeric (18,2) 
1	Mirae Asset Emerging Bluechip Fund - Regular - Growth	49046.00
2	Kotak Emerging Equity Fund - Regular - Growth	47469.00
3	Nippon India Small Cap Fund - Regular - Growth	43630.00
4	DSP Top 100 Equity Fund - Regular - Growth	41828.00
5	UTI Mid Cap Fund - Regular - Growth	41728.00

-- Query 2. average NAV per month

```
SELECT
d.year,
d.month_name,
ROUND(AVG(n.nav),2) AS avg_nav
FROM fact_nav n
JOIN dim_date d
ON n.date_id = d.date_id
GROUP BY
d.year,
d.month,
d.month_name
ORDER BY
d.year,
d.month;
```

	year integer 	month_name character varying (20) 	avg_nav numeric 
1	2022	January	207.06
2	2022	February	207.72
3	2022	March	209.69
4	2022	April	211.83
5	2022	May	212.73
6	2022	June	213.86
7	2022	July	213.96
8	2022	August	215.68
9	2022	September	218.49
10	2022	October	219.53
11	2022	November	223.47
12	2022	December	226.76
13	2023	January	230.67
14	2023	February	233.85
15	2023	March	238.01
16	2023	April	240.61
17	2023	May	241.89
18	2023	June	244.61
19	2023	July	245.78

-- Query 3. SIP Year-over-Year (YoY) Growth

```
SELECT
  d.year,
  SUM(t.amount_inr) AS sip_amount
FROM fact_transactions t
JOIN dim_date d
ON t.date_id = d.date_id
WHERE transaction_type = 'SIP'
GROUP BY d.year
ORDER BY d.year;
```

	year integer 	sip_amount numeric 
1	2024	61026046.00
2	2025	26279263.00

-- Query 4. Transactions by State

```
SELECT
  state,
  COUNT(*) AS total_transactions,
  SUM(amount_inr) AS total_amount
FROM fact_transactions
GROUP BY state
ORDER BY total_amount DESC;
```

	state character varying (100) 	total_transactions bigint 	total_amount numeric 
1	Tamil Nadu	1100	126020641.00
2	Madhya Pradesh	1137	124230586.00
3	West Bengal	1130	121974095.00
4	Punjab	1164	121295845.00
5	Uttar Pradesh	1076	119497728.00
6	Gujarat	1113	117945641.00
7	Delhi	1083	116939568.00
8	Rajasthan	1010	114352455.00
9	Karnataka	1057	113816812.00
10	Haryana	1096	112280777.00
11	Telangana	1092	111713771.00
12	Maharashtra	990	101346271.00

--Query 5. Funds with Expense Ratio < 1%

```

SELECT
    scheme_name,
    fund_house,
    expense_ratio_pct
FROM dim_fund
WHERE expense_ratio_pct < 1
ORDER BY expense_ratio_pct;

```

	scheme_name character varying (250) 🔒	fund_house character varying (150) 🔒	expense_ratio_pct numeric (5,2) 🔒
1	Nippon India Gilt Securities Fund - Regular - Growth	Nippon India MF	0.55
2	HDFC Short Term Debt Fund - Regular - Growth	HDFC Mutual Fund	0.56
3	Kotak Liquid Fund - Regular - Growth	Kotak Mahindra MF	0.60
4	SBI Bluechip Fund - Direct Plan - Growth	SBI Mutual Fund	0.66
5	SBI Small Cap Fund - Direct Plan - Growth	SBI Mutual Fund	0.72
6	Nippon India Large Cap Fund - Direct - Growth	Nippon India MF	0.72
7	ICICI Pru Liquid Fund - Regular - Growth	ICICI Prudential MF	0.74
8	Axis Bluechip Fund - Direct - Growth	Axis Mutual Fund	0.75
9	SBI Magnum Gilt Fund - Regular Plan - Growth	SBI Mutual Fund	0.77
10	HDFC Mid-Cap Opportunities Fund - Direct - Growth	HDFC Mutual Fund	0.78
11	ABSL Liquid Fund - Regular - Growth	Aditya Birla Sun Life MF	0.79
12	ICICI Pru Bluechip Fund - Direct - Growth	ICICI Prudential MF	0.80
13	Nippon India ETF Nifty 50 BeES	Nippon India MF	0.89
14	HDFC Top 100 Fund - Direct Plan - Growth	HDFC Mutual Fund	0.92

--Query 6. Top 10 Funds by 5-Year Return

```

SELECT
    f.scheme_name,
    p.return_5yr_pct
FROM fact_performance p
JOIN dim_fund f
ON p.amfi_code = f.amfi_code
ORDER BY p.return_5yr_pct DESC
LIMIT 10;

```

	scheme_name character varying (250) 🔒	return_5yr_pct numeric (8,2) 🔒
1	ABSL Small Cap Fund - Regular - Growth	23.80
2	Axis Small Cap Fund - Regular - Growth	22.62
3	Nippon India Small Cap Fund - Regular - Growth	21.88
4	SBI Small Cap Fund - Direct Plan - Growth	21.82
5	SBI Small Cap Fund - Regular Plan - Growth	20.67
6	DSP Small Cap Fund - Regular - Growth	20.61
7	DSP Midcap Fund - Regular - Growth	19.00
8	Axis Midcap Fund - Regular - Growth	18.94
9	Kotak Emerging Equity Fund - Regular - Growth	17.75
10	HDFC Mid-Cap Opportunities Fund - Regular - Growth	17.69

--Query 7. Transaction Type Distribution

```

SELECT
    transaction_type,
    COUNT(*) AS total_transactions,
    SUM(amount_inr) AS total_amount
FROM fact_transactions
GROUP BY transaction_type
ORDER BY total_transactions DESC;

```

	transaction_type character varying (20) 🔒	total_transactions bigint 🔒	total_amount numeric 🔒
1	SIP	7824	87305309.00
2	Lumpsum	3217	813153402.00
3	Redemption	2007	500955479.00

--Query 8. Fund Count by Category

```

SELECT
    category,
    COUNT(*) AS total_funds
FROM dim_fund
GROUP BY category
ORDER BY total_funds DESC;

```

	category character varying (100) 🔒	total_funds bigint 🔒
1	Equity	34
2	Debt	6

--Query 9. Average Investment by City Tier

```

SELECT
    city_tier,
    ROUND(AVG(amount_inr),2) AS avg_investment
FROM fact_transactions
GROUP BY city_tier
ORDER BY avg_investment DESC;

```

	city_tier character varying (20) 🔒	avg_investment numeric 🔒
1	B30	109371.73
2	T30	106415.94

--Query 10. Top Fund Houses by Number of Schemes

```

SELECT
    fund_house,
    COUNT(*) AS total_schemes
FROM dim_fund
GROUP BY fund_house
ORDER BY total_schemes DESC;

```

	fund_house character varying (150) 	total_schemes bigint 
1	Nippon India MF	5
2	HDFC Mutual Fund	5
3	ICICI Prudential MF	5
4	SBI Mutual Fund	5
5	Kotak Mahindra MF	4
6	Axis Mutual Fund	4
7	DSP Mutual Fund	3
8	Aditya Birla Sun Life MF	3
9	UTI Mutual Fund	3
10	Mirae Asset MF	3